

Analytical Problem Solving

SOLUTION

Question 1 – Solution

Solution

The Water Jug Problem is solved using **Breadth-First Search (BFS)**, which explores the state space level by level until the goal state is reached.

Each state is represented as (x, y) , where:

- x = Amount of water in the 4-gallon jug
- y = Amount of water in the 3-gallon jug

Initial State: $(0,0)$

Goal State: $(2,0)$

State Space Solution

Step	State (4G,3G)	Action
1	$(0,0)$	Initial empty state
2	$(0,3)$	Fill the 3-gallon jug
3	$(3,0)$	Pour water from the 3-gallon jug into the 4-gallon jug
4	$(3,3)$	Fill the 3-gallon jug again
5	$(4,2)$	Pour water into the 4-gallon jug until it becomes full
6	$(0,2)$	Empty the 4-gallon jug
7	$(2,0)$	Transfer the remaining 2 gallons into the 4-gallon jug (Goal State)

Result: The goal state $(2,0)$ is successfully reached.

Question 2 – Solution

Solution

(i) Percepts

The Mars Rover receives information from:

- Cameras
- Rock and soil sensors
- Temperature sensors
- Atmospheric pressure sensors
- Obstacle detection sensors
- Battery level sensor
- Position (GPS/Odometry)

(ii) Operating Environment

Property	Description
Observability	Partially Observable
Agents	Single Agent
Deterministic	Stochastic
Episodic	Sequential
Environment	Dynamic
State Space	Continuous
Knowledge	Unknown

(iii) Actions

The Mars Rover can:

- Move forward
- Move backward

- Turn left
- Turn right
- Capture images
- Collect soil samples
- Analyze rocks
- Avoid obstacles
- Send data to Earth
- Recharge using solar panels

(iv) Performance Measure

The performance is evaluated based on:

- Number of samples collected
- Accuracy of analysis
- Safe navigation
- Energy efficiency
- Communication success
- Mission completion

(v) Suitable Agent Architecture

Model-Based Goal-Based Learning Agent

Reason:

The Mars Rover maintains an internal model of its environment, plans paths, learns from previous experiences, and works toward achieving mission goals.

Question 3 – Solution

Solution

The **8-Queens Problem** is solved using the **Backtracking Search Algorithm**, which places one queen at a time and removes it if a conflict occurs.

Problem Components

Initial State:

- Empty 8×8 chessboard

State Space:

- All possible arrangements of 8 queens

Operators:

- Place one queen in a safe position in each column

Goal Test:

- No two queens are in the same row, column, or diagonal

Path Cost:

- Each queen placement has equal cost

Search Strategy

- Constraint Satisfaction Problem (CSP)
- Backtracking Search
- Depth-First Search (DFS)

One Valid Solution

```
Q . . . . . . .
. . . . Q . . .
. . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . Q .
. Q . . . . .
. . . Q . . . .
```

Result: All eight queens are placed without attacking each other.

Question 4 – Solution

Solution

The OLA Cab Booking System is modeled using a **Goal-Based Problem-Solving Agent**.

Agent Type

Goal-Based Agent

Reason

The agent selects the best cab and route to achieve the user's goal of reaching the destination safely and efficiently.

Pseudocode

START

Input Pickup Location

Input Destination

Display Available Cab Types

User Selects Cab

Search Available Drivers

IF Driver Available THEN

 Calculate Fare

 Confirm Booking

 Assign Driver

 Navigate to Destination

ELSE

 Display "No Cab Available"

ENDIF

STOP

Result: The Goal-Based Agent successfully books the cab and completes the trip.

Question 5 – Solution

Solution

The delivery network is solved using the **Uniform Cost Search (UCS)** algorithm.

Step-by-Step UCS Execution

Step	Expanded Node	Cost
1	S	0
2	A	1
3	C	2
4	D	3
5	B	4
6	G	4

Least Cost Path

$S \rightarrow A \rightarrow C \rightarrow G$

Total Cost

$$1 + 1 + 2 = 4$$

Result

Using the **Uniform Cost Search (UCS)** algorithm, the optimal path from **S** to **G** is:

$S \rightarrow A \rightarrow C \rightarrow G$

Minimum Cost = 4